

Applied AI - Homework 1

Names:

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```
joeyan@jupyter-argueta-erik-csu-fullerton-edu---d93e1a9d:~$ python3 data_tester.py
=====
| Question 1(a) |
=====
Array Method:
Normalization time: 3.035930 seconds

Numpy Method:
Normalization time: 1.885677 seconds

Time difference: 1.150254 seconds

=====
| Question 1(b) |
=====
X training: (331761, 4)
Y training: (331761,)
X testing: (82941, 4)
Y testing: (82941,)

=====
| Question 2(a) & (b) |
=====
Python Largest Column: TC
Python time: 0.172918 seconds.
NumPy Largest Column: TC
NumPy time: 0.000894 seconds.
NumPy speedup: 193.32x
Time difference: 0.172024 seconds.

=====
| Question 2(c) & (d) |
=====
float32 matrix time: 0.001341 seconds
float64 matrix time: 0.001608 seconds
Time difference: 0.000266 seconds
Result shape: (4, 4)

=====
| GPU and CPU Comparison|
=====
GPU: NVIDIA A10
Data type: torch.float64
Input shape: (414702, 4)
CPU average time: 0.001138800 seconds
GPU average time: 0.002037742 seconds
Time difference: 0.000899 seconds.
GPU speedup: 0.56x
CPU performance: 11.653 GFLOPS
GPU performance: 6.512 GFLOPS
joeyan@jupyter-argueta-erik-csu-fullerton-edu---d93e1a9d:~$
```

1a. Numpy performs normalization faster than Python's standard array functions; it performs about 2.161690 seconds

2b. Numpy performs dot products faster than using a loop; Numpy computed the dot product 0.184421 seconds faster.

2c. There is a difference between 32-bit and 64-bit computation times. A likely explanation is that 32-bit occupies less memory than 64-bit, thus less time to compute.

2d. Hardware Specifications:

- CPU Clock Speed: AMD EPYC 7502 32-Core Processor - max 2.5GHz
- # of CPU Cores: 64
- System RAM size: 503 GB
- GPU Model: Nvidia A10
- GPU Clock Speed: 1,695 MHz max
- # of GPU Cores: 9,216
- GPU Memory size: 24 GB GDDR6
- GPU Bandwidth: 600 GB/s

The GPU has a theoretical peak performance of 31.2 FLOPS using FP32, and an approximate theoretical performance of 0.488 TFLOPS. The matrix multiplication requires approximately 13.27 million operations, with an ideal time of approximately 27.2 microseconds. Factors like memory access and overhead tasks will slow the actual execution time.

CPU Specs

```
joyvan@jupyter-argueta-erik-csu-fullerton-edu---d93e1a9d:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          43 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 128
On-line CPU(s) list:    0-127
Vendor ID:              AuthenticAMD
Model name:             AMD EPYC 7502 32-Core Processor
CPU family:             23
Model:                  49
Thread(s) per core:     2
Core(s) per socket:     32
Socket(s):              2
Stepping:               0
Frequency boost:        enabled
CPU(s) scaling MHz:     112%
CPU max MHz:            2500.0000
CPU min MHz:            1500.0000
BogoMIPS:               5000.10
```

RAM Specs:

```
joyvan@jupyter-argueta-erik-csu-fullerton-edu---d93e1a9d:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          503Gi        65Gi        140Gi         2.3Gi        303Gi        437Gi
Swap:          0B           0B           0B
```

(GPU Specs next page)

GPU Specs:

```
Timestamp                : Mon Sep 14 03:25:36 2026
Driver Version            : 595.71.05
CUDA Version              : 13.2

Attached GPUs             : 1
GPU 00000000:81:00.0
  Product Name            : NVIDIA A10
  Product Brand           : NVIDIA
  Product Architecture    : Ampere
  Display Mode            : Requested functionality has been deprecated
  Display Attached        : Yes
  Display Active          : Disabled
  Persistence Mode        : Enabled
  Addressing Mode         : HMM
```

```
Clocks
  Graphics                : 210 MHz
  SM                      : 210 MHz
  Memory                  : 405 MHz
  Video                   : 555 MHz
```

```
Max Clocks
  Graphics                : 1695 MHz
  SM                      : 1695 MHz
  Memory                  : 6251 MHz
  Video                   : 1575 MHz
Max Customer Boost Clocks
  Graphics                : 1695 MHz
```